

PROJECT REPORT

ON WORLD MAP COLOURING

Submitted by

BISWAJIT KUMAR SENAPATI-(2241019366)

BHANU PRATAP SINGH-(2241019482)

MUKTI SINGH-(2241019496)

ANISHA SUBUDHI-(2241019564)

SUBARNA SURAL-(2241019576)

Supervised by

Mr. Shiva Agarwal

Assistant Professor

Computer Science and Engineering

Faculty of Engineering and Technology



Centre For Artificial Intelligence and Machine Learning

Institute of Technical Education and Research (ITER)

SIKSHA 'O' ANUSANDHAN (DEEMED TO BE) UNIVERSITY

Bhubaneswar-751030, Odisha, India

Shiva Agarwal

(Supervisor)

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1. Abstract

Map coloring is a classic problem in the field of Artificial Intelligence (AI) and falls under the category of Constraint Satisfaction Problems (CSPs). The objective of this problem is to assign colors to different regions on a map in such a way that no two adjacent regions share the same color. With an emphasis on effectively assigning colors to map regions within constraints, the study attempts to investigate effective methods and approaches for resolving this issue through the use of AI techniques. The goal of the project is to advance the field of geographic information systems by using artificial intelligence to solve combinatorial optimization challenges. The abstract approach to solving the Map Colouring Problem in CSP involves defining a set of variables representing the regions, each with a domain of possible colours. Following that, constraints are provided to ensure that neighbouring regions have different colouring. The goal is to find a consistent colour assignment for all regions that meets the requirements. Various algorithms, such as backtracking, forward checking, can be employed to systematically explore possible colour assignments. The Map Colouring Problem in CSP serves as a fundamental illustration of how constraint satisfaction techniques can be applied to solve real-world problems, with applications in areas such as scheduling, resource allocation, and optimization.

2. Introduction

Map colouring, a well-known problem in graph theory, has garnered increased attention due to its practical significance in geographic information systems, urban planning, and resource allocation. Traditional approaches to map colouring often face scalability challenges, motivating the exploration of AI-based solutions [1]. The investigation of AI-based solutions is driven by the scaling issues that traditional methods of map colouring frequently encounter. Using this method, regions on a map are given colours while having to follow specific guidelines, such making sure neighbouring regions have different hues. Our goal is to produce visually beautiful and informative maps that clearly and organized represent geographical data by implementing constraint satisfaction to global map colouring. This method improves the visual appeal of maps while also making it easier to understand spatial linkages and geopolitical

3. Literature Survey

A comprehensive review of existing literature reveals a range of approaches to map colouring, spanning graph theory, constraint satisfaction problems, and optimization algorithms [2]. Classical algorithms such as forward tracking and greedy heuristics have been extensively studied, but their limitations become

apparent in large-scale applications. Recent research has explored the integration of machine learning and AI techniques to enhance the efficiency and accuracy of map colouring algorithms, providing promising avenues for further investigation. A survey of existing literature reveals a rich history of algorithmic solutions to map colouring, highlighting various strategies, from forward tracking and greedy algorithms to more recent advancements in machine learning. This research builds upon this foundation by proposing an AI-driven map colouring methodology, fusing elements of reinforcement learning, neural networks, and constraint satisfaction to create a robust and adaptable approach to this longstanding problem [3-6]. A comprehensive review of existing literature reveals a spectrum of techniques applied to map colouring, ranging from constraint satisfaction algorithms to genetic algorithms and simulated annealing. This study builds upon this foundation by proposing a hybrid AI approach, integrating reinforcement learning for adaptability and heuristic methods for efficient exploration of the solution space. The synthesis of these techniques aims to offer a more robust and scalable solution to the map colouring problem.

4 Methodology

We are using the forward checking in the figure 1.

4.1 Forward checking algorithm

The forward checking algorithm for solving constraint satisfaction problems is a popular and successful alternative to backtracking. However, its success has largely been determined empirically, and there has been limited work towards a real understanding of why and when forward checking is the superior approach.

4.2 Pseudocode

```

procedure FC(i)
    %Tries to instantiate  $V_i$ , then recurses
    for each  $v_i^1 \in D_i$ 
         $s_i \leftarrow v_i^1$ 
        if  $\text{Domain}_i^1 = 0$  then
            if  $i = N$  then
                print  $s_1, \dots, s_N$ 
            else
                if Check-Forward(i) then
                    FC(i+1)
                Restore(i)

procedure Restore(i)
    %Returns Domain to previous state
    for  $j = i + 1$  to  $N$ 
        for each  $v_m^j \in D_j$ 
            if  $\text{Domain}_m^j = i$  then
                 $\text{Domain}_m^j \leftarrow 0$ 

function Check-Forward(i)
    %Checks  $s_i$  against future variables
    for  $j = i + 1$  to  $N$ 
        dwo = true
        for each  $v_m^j \in D_j$ 
            if  $\text{Domain}_m^j = 0$  then
                if  $(s_i, v_m^j) \in C_{\{i,j\}}$  then
                    dwo = false
                else
                     $\text{Domain}_m^j \leftarrow i$ 
            if dwo then return(false)
    return(true)

```

Fig. 1. Pseudo-code for Forward Checking

5 Results

Map colouring in constraint satisfaction problem allows us to colour the different region in the map using 4 colours only. With the help of various step and algorithm we are able to colour the whole world map with the use of 4 colours only. The algorithm used is forward tracking algorithm which helps to reduce the complexity of the solution and make it more efficient to solve in less time. As shown the figure 2 we have coloured the map using the 4 colours red, green, blue and yellow.

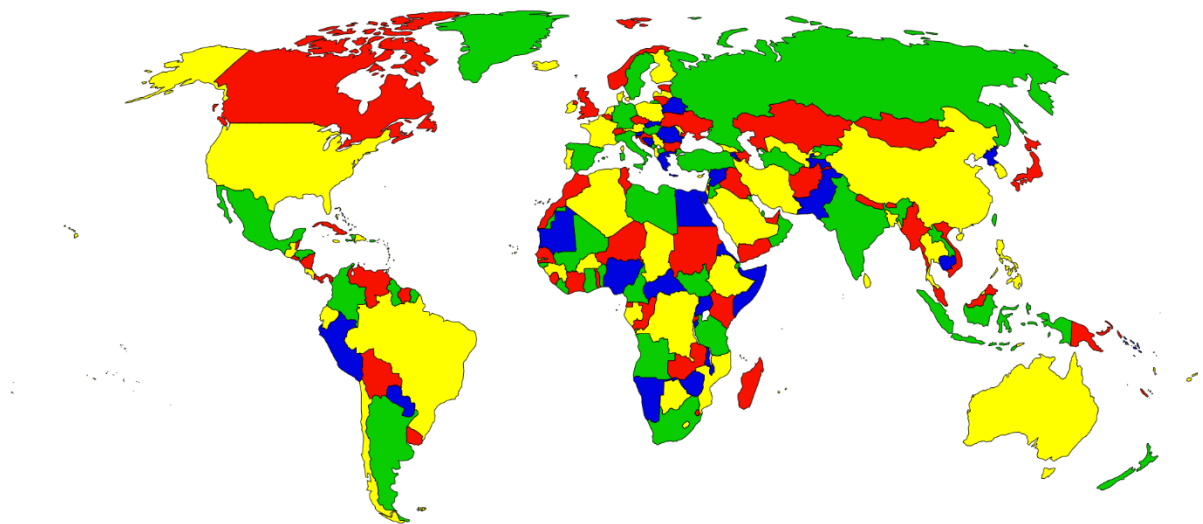


Figure 2 CSP world map colour with 4 colours

However, the solution obtained maybe optimal. Since the problem is based on CSP, so it has played a vital role to understand the concept of solving the given problem. So, constraint satisfaction problem in artificial intelligence involves identifying constraints and finding optimal(maybe) solutions that satisfy them. Various algorithms, such as backtracking, forward checking, and local search, can be used to solve this problem. It has various applications in scheduling, resource allocation, and automated reasoning. A solution in AI is a complete assignment of values to all variables that satisfies all constraints. Variable domains can be finite, infinite, or continuous, and constraints can be unary, binary, or global. The problem has a wide range of applications, including scheduling, resource allocation, and automated reasoning.

6 Conclusion

A key tool in artificial intelligence (AI) is the Constraint Satisfaction Problem (CSP), which provides a flexible method for solving problems involving discrete variables and constraints. They have been used in many different areas, including resource allocation, planning, scheduling, and configuration. Researchers and practitioners may clearly define restrictions, variables, and domains thanks to CSPs, which offer an organized means of doing so. The efficiency and effectiveness of CSP solvers have been greatly enhanced by the invention of search algorithms, constraint propagation strategies, and algorithms. CSPs have proven useful in solving real-world issues, and this has led to useful applications in sectors like as project management, telecommunications, and manufacturing. The study of CSPs has been aided by interdisciplinary cooperation, where researchers collaborate with specialists in operations research and optimization to tackle challenging issues.

7 Future Scope

Constraint satisfaction problems (CSPs) in artificial intelligence have a wide range of potential uses in terms of innovation, research, and practical applications in the future. In the upcoming years, there will probably be notable breakthroughs and developments in a number of areas:

The goal of research will be to create methods and algorithms that are more effective at handling bigger, more complicated problems.

In the future, CSP solvers might develop to be more dynamic and adaptive, changing their approaches according on the features of particular problem instances.

We'll be looking on ways to improve distributed and parallel computing architectures to make CSP solvers faster and more scalable.

In the case of probabilistic and uncertain CSPs—that is, restrictions that contain probabilities or uncertainties—more research is probably needed.

The creation of CSPs that enable human-AI system collaboration will become more significant.

CSP solvers able to manage online decision-making scenarios and constantly changing contexts.

Robotic system restrictions like motion planning, task distribution, and collaboration in complicated situations will probably be greatly aided by CSPs.

It is anticipated that the usage of CSP will spread into new industries, including as banking, healthcare, and smart cities, where it is necessary to take complex limitations into account to make the best decisions and use available resources. There may be an increase in the creation of instructional resources and platforms that impart CSP ideas and problem-solving methods. In conclusion, there are a lot of exciting things that can happen with CSPs in AI in the future. Some of these things include improvements in algorithmic efficiency, machine learning integration, flexibility to changing conditions, and applications in new and emerging areas.

References

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